Tshwane University of Technology

We empower people

DIPLOMA IN COMPUTER SCIENCE (EXTENDED)

WORK INTEGRATED LEARNING

RULES AND OUTCOMES

WIL MODULE 1: WIL IN COMPUTER SCIENCE EXT WOC316D

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WIL MODULE : WIL IN COMPUTER SCIENCE EXT WOC316D

RULES

QUALIFICATION TITLE :	DIPLOMA IN COMPUTER SCIENCE (EXTENDED)
WIL MODULE :	WIL IN COMPUTER SCIENCE EXT WOC316D
WIL MODULE CODE :	WOC316D
WIL MODULE SUBJECT :	WORK-INTEGRATED LEARNING

DURATION OF WIL MODULE	6 MONTHS
WIL CREDITS	60 CREDITS
NUMBER OF ON SITE VISITATION	1
ASSESSMENT METHODOLOGY	OUTCOMES BASED ASSESSMENT

PREREQUISITE SUBJECTS TO REGISTER

The student must pass the following prerequisite Subjects to qualify to Register for WIL IN COMPUTER SCIENCE EXT WOC316D

1st year		
2nd year		
3rd year		
4th year		
	INTERNET PROGRAMMING	MOBILE COMPUTING
	SOFTWARE PROJECT	

WELCOME

Dear Student

Welcome to a very interesting phase of your studies.

By now you are in a fortunate position to be placed at a workstation for this specific level of your work-integrated Learning.

As you probably know by now, finding a position at a workstation is not easy as the requests for placement exceed the placement possibilities by far. So do your utmost and give your full co-operation to your mentor (supervisor). You still have a lot to learn, and your future depends on your attitude towards your vocation.

Please read through all the information supplied and make sure that you understand fully what is expected of you. Your progress depends on you and will be reflected in your academic record.

Please use your tut4life email address for all correspondence regarding your academic matters. Make sure you have access to your tut4life emails at all times.

Last but not least you and your mentor are not alone in this venture. We are there to support and guide you both.

So, should you need any help (including problems, advice, or enquiries), do not hesitate to contact your Cooperative Education Lecturer, WIL Coordinator, or the Office of Co-operative Education.

INTRODUCTION

The purpose of this study guide is to provide the learner with the information needed that is crucial in completing this module. This study guide serves as a contract between the learner and the lecturer regarding what is to be done, and on which date it should be completed. All rules and regulations stipulated in this study guide should be aligned with the standard rules and regulations of Tshwane University of Technology. Any deviation from these standard rules and regulations will be regarded as invalid and must be reported to the HOD of the module's custodian department.

The WIL module comprises of three modalities:

MODE 1: PREPARATION FOR INDUSTRY (PI) / MENTOR EVALUATION

The **PREPARATION FOR INDUSTRY** modality involves formal theoretical learning of organisational behavior. The modality is carried out through Online learning where you (the student) are required to familiarise yourself with this approach as well as the content in this study guide to ensure you are aware of the requirements and expectations.

Students are required to download the **Mentor Evaluation form** via the TUT MyWIL portal documents menu. The Mentor Evaluation form is accessible under the "F**ORMS & TEMPLATES**" section. The Mentor Evaluation must be downloaded and handed to the Industry mentor to complete and sign off. Once the evaluation form is completed and signed, students must then submit the form via the TUT MyWIL Portal and link it to "**OUTCOME MENTOR EVALUATION**" for approval.

MODE 2: INDUSTRIAL EXPOSURE (IE)

The INDUSTRIAL EXPOSURE modality comprises six (6) months work integrated learning tasks/activities conducted by the student at a valid workstation and submitted via the TUT MyWIL portal. The WIL Coordinator / Lecturer will ensure that the six (6) months requirements for the Industrial Exposure period are met by conducting regular visitations (telephonic or physical) at the student's workstation.

MODE 3: INDUSTRY PROJECT (IP)

The INDUSTRY PROJECT modality comprises the project to be planned and executed using business analysis and project management techniques during the six-month WIL period. The project is split into two (2) phases.

The initial phase (three (3) months of the WIL period) involves a presentation of the analyzed existing business processes and a proposed effective solution with the proposed project plan. The final phase (the sixth (6) month of the WIL period) involves the presentation of the implemented solution and the results thereof.

PURPOSE OF THIS MODULE

This Work Integrated Learning module is a 60-credit module on NQF level 6 which prepares the student to integrate the application of the concepts and principles learned through the theoretical learning of all the modules in the qualification. The student will be able to apply his/her knowledge of the theory learned in the qualification. Upon completion of the module, the student will be able to analyze and behave professionally in the working environment

This module forms part of the qualification Diploma in Computer Science Extended and will individually contribute to achieving the exit level outcomes of:

- **ELO 1:** Design and analyze algorithms and data structures to create efficient.
- **ELO 2**: Develop Web and Mobile and Desktop and client server-based applications and solutions.
- **ELO 3**: Develop mobile device applications.
- **ELO 4**: Optimize, test, and maintain software applications on different platforms.
- **ELO 5**: Design and implement databases to support programming solutions.
- **ELO 6**: Apply software engineering principles to the application development.
- **ELO 7**: Apply software integration, verification, and validation techniques.
- **ELO 8**: Apply interaction design principles.
- **ELO 9:** Develop secure mobile and web-based applications.
- **ELO 10**: Apply software quality techniques (elective).
- **ELO 11**: Install and manage Web servers (elective).
- **ELO 12**: Apply Web content management techniques (elective).

This module is placed in the third year as it is a core module and will require a majority of the students' time, to develop their skills and knowledge in order to become junior: programmers or developers. Once a student completes this module, he/she will have a solid understanding of the working environment.

The NQF of this module is 6 and, on this level, it is expected that the learner should display the ability to apply understanding of basic concepts, facts, and terms, select and apply standard methods, procedures, and/or techniques, as well as design an information system, and compliance to the working environment.

The graduate attributes of the institution expect students to be well-rounded individuals who will demonstrate knowledge, skills, and applied competencies. By employing the proper outcomes, assessment methods, and level descriptors, this module will demonstrate the student's knowledge and skills such as adapting and conforming to the professional working environment.

The further intent of the module is to also ensure that the students are well-rounded, and this can only be ensured if the individuals can interact effectively with their changing environment and act responsive and responsible within a variety of social and cultural settings. Aligned to the graduate attributes and ELO's students will be guided through activities like problem-solving and reflexive practices. In addition, students will acquire the following: Knowledge: understand key terms, concepts, facts, principles, rules, and theories. Skills: Select and apply standard methods, procedures, and/or techniques Plan, manage, and implement processes within a supported environment. The module will require students to work individually and in groups thereby contributing towards both personal and professional ethics. As the students progress in their studies, they will be required to present their work to other fellow students, thereby developing their presentation and communication skills (skills identified as graduate attributes by

the university). The technical knowledge obtained in this module together with the communication skills and presentation skills will prepare the students for the workplace. This ties in with the instructional goal - To prepare diverse students for rewarding careers and responsible citizenry by providing a student-centered learning experience that is underpinned by a scholarship of teaching and learning

CODE OF CONDUCT

- Students must register and follow the instructions and guidelines for WIL as set out in the prospectus.
- All issues pertaining to the module must be reported to your lecturer. If the issue cannot be resolved adequately it must be escalated to the module coordinator. If the issue is still not resolved, it must be escalated to the section head. If the issue is still not resolved, it can be escalated to the HOD of the module's custodian department. Students may only report issues to the Dean of the faculty if the issue cannot be resolved by the HOD. Under no circumstances will issues be addressed if the proper reporting chain is not followed.

LEARNER	LECTURER	MODULE	CORDINATOR	SECTION HEAD	HOD	DEAN	DVC

CONTACT INFORMATION OF CUSTODIAN DEPARTMENT OF THIS MODULE

The custodian department of this module is the Department of Computer Science.

Name & Surname	Campus	Office Location	Contact number and Email	Role in Programme
Ms. Molly Moche	Soshanguve South	**20-103**	012 382 9938 mollymoche@tut.ac.za	**Departmental Administrator**
Ms. Sinnah Mokhutso	Soshanguve South	**20-G05**	012 382 9631 mokhutsoms@tut.ac.za	**Postgraduate Departmental Administrator**
Ms. Angelina Shonaphi Bhembe	Emalahleni	**14-G37**	013 653-3165 bhembeas@tut.ac.za	**Administrator**
Ms. Maesela Lebelo	Polokwane	**1-G247**	015 287-0757 lebelom@tut.ac.za	**Administrator**

CONTACT INFORMATION OF LECTURERS

This module has one primary module coordinator who is responsible for coordinating the activities of the module.

If this module is offered at one of the other learning sites and the primary module coordinator does not reside on the learning site, the activities of this module are managed by the primary module coordinator via a site coordinator. There is one facilitator assigned to facilitate the teaching of each group.

Module coordinator/s

Module Coordinator	Dr. Daniel Chuene
Office Telephone Number	012 382 9267
Email address	ChueneND@tut.ac.za

Module Coordinator	Mr. Vuyisile Memani
Office Telephone Number	012 382 9749
Email address	MemaniV@tut.ac.za

Module Coordinator	Mr. Craven Klaas Lepota
Office Telephone Number	012 382 9014
Email address	LepotaCK@tut.ac.za

Module Coordinator	Mr. Thanyani Netshisumbewa
Office Telephone Number	012 382 9843
Email address	NetshisumbewaT@tut.ac.za

Module Coordinator	Mrs. Rejoice Mokhari
Office Telephone Number	012 382 9246
Email address	MokhariMR@tut.ac.za

Module Coordinator	Mrs. Vivian Masuku
Office Telephone Number	012 382 9854
Email address	MasukuMV@tut.ac.za

Module Coordinator	Mrs. Nokubongwa Zuma
Office Telephone Number	012 382 9012
Email address	ZumaNF@tut.ac.za

Module Coordinator	Mr. Solomon Adeyemi Odunaike

Office Telephone Number	012 382 9151
Email address	OdunaikeSA@tut.ac.za

LIBRARY CONTACT DETAILS

Soshanguve Faculty Librarian	Ms Esme Refilwe Nkosi
Office Location	Building 6, Room 105 (in the library)Building 6, Room 105 (in the library)
Office Telephone Number	012 382 6160
Email address	nkosierl@tut.ac.za

CONSULTATION

CONSULTATION AND AVAILABILITY OF LECTURER(S)

Students seeking appointments or to consult with lecturers can now conveniently utilize the Chatroom feature available on the MyWIL Portal. Through this platform, students can schedule face-to-face appointments, communicate concerns, or request assistance directly from the lecturer.

To initiate a conversation or book an appointment (where required), simply access the Chatroom on the MyWIL Portal. This serves as a tool for scheduling appointments and facilitating effective communication between students and lecturers.

Please remember to mention the topic to be discussed during the consultation session.

MODERATION

This is a final year module; therefore, moderation will be done externally by a moderator approved by Senate. The moderator will be competent in the area of corporate governance, ethics and risk management including the field of auditing and/or internal auditing.

PLAGIARISM

The following is an extract from TUT's plagiarism policy. **Reference: RIPPOL067.**

All students have a moral obligation and responsibility to maintain the following academic integrity principles in the production and presentation of academic outputs, regardless of the presentation format and/or work type:

Each student should only submit his/her own original academic work, except when formal group work is required in the production of the academic output;

Each student should accurately indicate in all academic outputs when information is used that was produced by another scholar by referencing it in accordance with a recognised referencing convention system;

No student should use, present, or submit someone else's electronic works, multimedia products or artistic works as if it is his/her own;

Each student should accurately indicate the download/access date and the uniform resource locator (URL) of the internet web page when information is used from a website, web page, or other electronic source;

No student should allow another person/s to use or copy from his/her academic output and present it as their own work;

Each student is required to attach a signed Declaration of Originality (or each academic output submission (e.g. assignment, project, manuscript, dissertation, and thesis); and Each student has the responsibility to request assistance from staff members should they require guidance and/or advice about plagiarism in their academic outputs.

Students have a moral obligation to report plagiarism incidents in academic and/or research environments. All whistle-blowers are protected in terms of the Policy on Prevention of Fraud, Corruption and Theft (Policy #: VCPOL010).

Plagiarism is a form of misconduct. The relevant part (Chapter 15-Student Discipline) of the Prospectus, Part 1 (Students' rules and regulations) read as follows:

"Any student who contravenes the provisions of rule 15.1 of the disciplinary code is guilty of misconduct and will be dealt with in terms of the disciplinary code for students 15.1.16 Handing in any written assignment for assessment in which the essential parts of the assignment have been copied from the work of another person, or any form of plagiarism."

All students must be fully aware that plagiarism offences/penalties can seriously affect their academic status and progress at TUT and other tertiary institutions. In the most serious cases, it can result in dismissal from the University and/or formal cancellation/retraction of current/previously submitted academic outputs. In addition, the University may indicate the nature and outcome of all plagiarism offences/penalties when it is required to provide a reference or conduct statement for the particular student.

PROMOTIONAL REQUIREMENTS

The following table reflects the allocation of marks for the calculation of the final mark on the TUT MyWIL portal;

Assessment	Code	Weight
Mentor Evaluation	n/a	25%
Industrial Exposure (IE) (monthly reports & final report)	IE	35%
Industry Project (IP)	IP	40% (Subminimum 40%)
TOTAL		100%

- The weights of the various marks may change during the six (6) months WIL period; these will be discussed with you when such arises.
- Marks will be captured continuously on ITS and will reflect on the TUT MyWIL portal once assessments are completed. If you have queries about your mark, you must immediately consult your subject lecturer (refer contact details).
- Once the Assessment mark is entered on TUT's mainframe computer, the marks cannot be changed.
- The pass mark for this module is 50%.

To pass the module a student must achieve a final mark of 50%.

SUMMARY OF OUTCOMES

#	OUTCOME	CORE OR ELECTIVE	CREDIT VALUE
1.0	INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 1 - MONTH 1	CORE OUTCOME	3
2.0	INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 2- MONTH 2	CORE OUTCOME	3
3.0	INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 3- MONTH 3	CORE OUTCOME	3
4.0	INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 4 - MONTH 4	CORE OUTCOME	3
5.0	INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 5 - MONTH 5	CORE OUTCOME	3
6.0	INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 6 - MONTH 6	CORE OUTCOME	3
7.0	INDUSTRY PROJECT (IP)	CORE OUTCOME	24
8.0	REPORT	CORE OUTCOME	3
9.0	MENTOR EVALAUTION	CORE OUTCOME	15
TOTAL CREDITS			60

1.0 : INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 1 - MONTH 1

DURATION OF TRAINING GUIDELINE	1 MONTH
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	5.00%
CREDIT VALUE	3
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

INSTRUCTIONS AND GUIDELINES TO STUDENT

Monthly Logsheet of Tasks/ Activities Completed.

The student is required to complete a monthly task/ activity report which must be submitted via the TUT MyWIL portal. The Task/ Activity report template is accessible via the TUT MyWIL portal's Documents Menu, located under the Forms and Templates section.

Monthly Log sheet templates must be downloaded from the Documents Menu>Forms & Templates section. Students can complete and submit in PDF format.

Shown below is an example of the template to be used for: Industrial Exposure (IE) - TASK & ACTIVITY REPORT MONTH 1 to REPORT MONTH 6

Industrial Exposure (IE) - TASK & ACTIVITY REPORT: LOG SHEET					
MONTH:					
FROM:					
TO:					
Summary of Tasks	**Duration**		**Evaluation**		
	Weeks	**Days**	**Poor**	**Satisfactory**	**Good**

Subtotal:	**Weeks**		**Days**		**Total to date:**	**Weeks**		**Days**	
Number of days absent from work:		**Reason**:							

Mentor/Supervisor Name & Signature: _____

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA	TYPE

COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

2.0 : INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 2- MONTH 2

DURATION OF TRAINING GUIDELINE	1 MONTH
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	5.00%
CREDIT VALUE	3
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

INSTRUCTIONS AND GUIDELINES TO STUDENT

Monthly Logsheet of Tasks/ Activities Completed.

The student is required to complete a monthly task/ activity report which must be submitted via the TUT MyWIL portal. The Task/ Activity report template is accessible via the TUT MyWIL portal's Documents Menu, located under the Forms and Templates section.

Monthly Log sheet templates must be downloaded from the Documents Menu>Forms & Templates section. Students can complete and submit in PDF format.

Shown below is an example of the template to be used for: Industrial Exposure (IE) - TASK & ACTIVITY REPORT MONTH 1 to REPORT MONTH 6

Industrial Exposure (IE) - TASK & ACTIVITY REPORT: LOG SHEET						
MONTH:						
FROM:						
TO:						
Summary of Tasks	**Duration**		**Evaluation**			
	Weeks	**Days**	**Poor**	**Satisfactory**	**Good**	

Subtotal:	**Weeks**		**Days**		**Total to date:**	**Weeks**		**Days**	
Number of days absent from work:		**Reason**:							

Mentor/Supervisor Name & Signature: _____

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA	TYPE

COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

3.0 : INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 3- MONTH 3

DURATION OF TRAINING GUIDELINE	1 MONTH
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	5.00%
CREDIT VALUE	3
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

INSTRUCTIONS AND GUIDELINES TO STUDENT

Monthly Logsheet of Tasks/ Activities Completed.

The student is required to complete a monthly task/ activity report which must be submitted via the TUT MyWIL portal. The Task/ Activity report template is accessible via the TUT MyWIL portal's Documents Menu, located under the Forms and Templates section.

Monthly Log sheet templates must be downloaded from the Documents Menu>Forms & Templates section. Students can complete and submit in PDF format.

Shown below is an example of the template to be used for: Industrial Exposure (IE) - TASK & ACTIVITY REPORT MONTH 1 to REPORT MONTH 6

Industrial Exposure (IE) - TASK & ACTIVITY REPORT: LOG SHEET					
MONTH:					
FROM:					
TO:					
Summary of Tasks	**Duration**		**Evaluation**		
	Weeks	**Days**	**Poor**	**Satisfactory**	**Good**

Subtotal:	**Weeks**		**Days**		**Total to date:**	**Weeks**		**Days**	
Number of days absent from work:		**Reason**:							

Mentor/Supervisor Name & Signature: _____

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA	TYPE

COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

4.0 : INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 4 - MONTH 4

DURATION OF TRAINING GUIDELINE	1 MONTH
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	5.00%
CREDIT VALUE	3
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

INSTRUCTIONS AND GUIDELINES TO STUDENT

Monthly Logsheet of Tasks/ Activities Completed.

The student is required to complete a monthly task/ activity report which must be submitted via the TUT MyWIL portal. The Task/ Activity report template is accessible via the TUT MyWIL portal's Documents Menu, located under the Forms and Templates section.

Monthly Log sheet templates must be downloaded from the Documents Menu>Forms & Templates section. Students can complete and submit in PDF format.

Shown below is an example of the template to be used for: Industrial Exposure (IE) - TASK & ACTIVITY REPORT MONTH 1 to REPORT MONTH 6

Industrial Exposure (IE) - TASK & ACTIVITY REPORT: LOG SHEET					
MONTH:					
FROM:					
TO:					
Summary of Tasks	**Duration**		**Evaluation**		
	Weeks	**Days**	**Poor**	**Satisfactory**	**Good**

Subtotal:	**Weeks**		**Days**		**Total to date:**	**Weeks**		**Days**	
Number of days absent from work:		**Reason**:							

Mentor/Supervisor Name & Signature: _____

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA	TYPE

COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

5.0 : INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 5 - MONTH 5

DURATION OF TRAINING GUIDELINE	1 MONTH
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	5.00%
CREDIT VALUE	3
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

INSTRUCTIONS AND GUIDELINES TO STUDENT

Monthly Logsheet of Tasks/ Activities Completed.

The student is required to complete a monthly task/ activity report which must be submitted via the TUT MyWIL portal. The Task/ Activity report template is accessible via the TUT MyWIL portal's Documents Menu, located under the Forms and Templates section.

Monthly Log sheet templates must be downloaded from the Documents Menu>Forms & Templates section. Students can complete and submit in PDF format.

Shown below is an example of the template to be used for: Industrial Exposure (IE) - TASK & ACTIVITY REPORT MONTH 1 to REPORT MONTH 6

Industrial Exposure (IE) - TASK & ACTIVITY REPORT: LOG SHEET					
MONTH:					
FROM:					
TO:					
Summary of Tasks	**Duration**		**Evaluation**		
	Weeks	**Days**	**Poor**	**Satisfactory**	**Good**

Subtotal:	**Weeks**		**Days**		**Total to date:**	**Weeks**		**Days**	
Number of days absent from work:		**Reason**:							

Mentor/Supervisor Name & Signature: _____

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA	TYPE

COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

6.0 : INDUSTRIAL EXPOSURE (IE) - TASK & ACTIVITY REPORT 6 - MONTH 6

DURATION OF TRAINING GUIDELINE	1 MONTH
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	5.00%
CREDIT VALUE	3
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

INSTRUCTIONS AND GUIDELINES TO STUDENT

Monthly Logsheet of Tasks/ Activities Completed.

The student is required to complete a monthly task/ activity report which must be submitted via the TUT MyWIL portal. The Task/ Activity report template is accessible via the TUT MyWIL portal's Documents Menu, located under the Forms and Templates section.

Monthly Log sheet templates must be downloaded from the Documents Menu>Forms & Templates section. Students can complete and submit in PDF format.

Shown below is an example of the template to be used for: Industrial Exposure (IE) - TASK & ACTIVITY REPORT MONTH 1 to REPORT MONTH 6

Industrial Exposure (IE) - TASK & ACTIVITY REPORT: LOG SHEET						
MONTH:						
FROM:						
TO:						
Summary of Tasks	**Duration**		**Evaluation**			
	Weeks	**Days**	**Poor**	**Satisfactory**	**Good**	

Subtotal:	**Weeks**		**Days**		**Total to date:**	**Weeks**		**Days**	
Number of days absent from work:		**Reason**:							

Mentor/Supervisor Name & Signature: _____

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA	TYPE

COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

7.0 : INDUSTRY PROJECT (IP)

DURATION OF TRAINING GUIDELINE	6 MONTHS
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	40.00%
CREDIT VALUE	24
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

Assessment of the IP modality of this module will include a presentation of the initial requirements phase of the project, followed by a presentation of the deployed solution and results at the end of the six (6) months WIL period.

The INDUSTRY PROJECT modality comprises of the project to be planned and executed using business analysis and project management techniques during the six months WIL period.

The project is split into two (2) phases.

The initial phase (three (3) months of the WIL period) involves a presentation of the analysed existing business processes and a proposed effective solution with the proposed project plan.

The final phase (sixth (6) month of the WIL period) involves the presentation of the implemented solution and the results thereof.

Present the final project

- The final computer science project is demonstrated.
- Relevant documentation is submitted.
- Design solution is implemented.
- Test results are demonstrated.

INDUSTRY PROJECT TEMPLATE

Software Development students are required to propose a three tier system, which is a client-server application where the user interface, processing logic and data management functions are physically separated.

You are required to follow the given template, and consult where necessary.

Phase 1 (Proposal)

Task	Description
1.Name of the Project	• Name the Project
2.Domain Analysis	• Explain the general field of business • Show understanding of terminology/glossary being used • Show the general knowledge and understanding of the business environment • Tasks and procedures currently performed • Customers and users

	• Competing software • Similarities to other domains
3.Define the Problem	• Express the Difficulty you want to solve from the domain • Or Opportunity that will result in benefit or improved productivity or sales
4.Define the Scope	Narrow the scope by defining a more precise problem • Apply knowledge of Integrated Result Based Management (IRBM) to define the Inputs, Activities, Outputs, outcomes, and the Impact that the application will have on the community • Answer the following questions: • Assess: What is the current situation? • Think: What caused it? Who is involved? • Envision: What are we going to achieve? • Plan: How are we going to do it? With whom? When? With what resources?
5.Vision and Objectives	• Write the Vision and Objectives (according to SMART principles) of the project
6.Users of the System	• Indicate the users of the system and their roles
7.Mandatory Functions	• The system should be able to Add/Register, Delete/Remove and Update data in the database
8.Functional Requirements	• Write the aspect of what the proposed system must do, which contribute in solving the customer's problem and represents a negotiated agreement among stakeholders • What inputs and outputs should the system accept • What computations should the system perform • The timing and synchronization of events
9.Non-functional requirements	• Describe Authentication(login/ logout) • Describe Availability
10.Use Case	• Describe sequence of actions that a user performs in order to complete a given task as a key activity in requirements using a diagram • This should cover full sequence of steps from beginning to until the end of the task • Describe the user's interaction with the system and not computations performed by the system • And not actions a user does manually

11.Tools and Technologies to be used	• Indicate the tools you intend to use for the project (e.g. Java, PHP, ASP, etc.)

Phase 2 (Modelling with Classes)

Task	Description
1.Class Diagrams	
2.Sequence Diagram	• Visualize how the system runs • Built from use case and class diagram
3.State Diagrams	• Describe the behavior of the system, activities and their transitions
4.Activity Diagrams	• Describe the flow of objects and components • Show representations of concurrent activities
5.Component Diagrams	
6.Deployment Diagram	

Phase 3 (User Interface)

Task	Description
1.Design User Interfaces	
2.Demo the Prototype	
3.Evaluate User Interface	• Use Heuristic Evaluation and report the possible usability defects
4.Validate Fields	• Verification and Validation

Phase 4 (Build the Database and Demonstrate Integration)

Task	Description
1.Build the database	• Define data structures
2.Manage objects	• Show objects, schema and integrity constraints
3.Normalization Process	• Normalize your database
4.Manipulate your data	• Populate your database using a script
5.Manage transaction	• Create transactions and database queries • This must correlate with functional requirements and use case

Phase 5 (Final Project Deliverance)

Task	Description
1.Application	• Fully working system • User Interface, Process Logic and Database integrated as a 3-tier system in a client server architecture
2.Document on Test cases and Test plan	• Compile a Test Case document and Test Plan for quality assurance on the system
3.Reports	• List reports to be generated by the system
4.Final Deliverable must include	• Application deployment execution and manual • Application archive (.war/.ear) with source code • Database backup and DDL Script • Complete Source code

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA		TYPE
NEW KNOWLEDGE ACQUIRED	Was the student able to give a thorough explanation of the new knowledge that he/she acquired?	0 TO 100%
PROBLEM SOLVING	Does the student demonstrate the ability to use or design algorithmic and structured thinking and problem decomposition and synthesis, etc.	0 TO 100%
APPLICATION OF KNOWLEDGE	Was the student able to give a thorough explanation of how he/she applied the old/new knowledge that he/she acquired?	0 TO 100%
CREATIVITY	Does the student demonstrate the ability to convert ill-defined problematic situations into software solvable problem?	0 TO 100%
MODELLING	Does the student demonstrate the ability to constructively create models such as Use Cases, Class Diagrams, Activity Diagrams or Sequence Diagrams?	0 TO 100%
DEBUGGING	Does the student demonstrate the ability to analyse the behaviour of code to diagnose a problem and find the underlying cause?	0 TO 100%
AWARENESS OF THE SOFTWARE DEVELOPMENT LIFE CYCLE	Does the student demonstrate an understanding of the types of methodologies (e.g. Agile or Waterfall) that can be used to progress through this cycle?	0 TO 100%
PROGRAMMING LEVEL DIFFICULTY	Does the student demonstrate the ability to read and write programs at different levels of difficulty during the industry exposure?	0 TO 100%
TECHNOLOGIES UNDERSTANDING	Does the student demonstrate the ability to understand the purpose of the technologies that he/she used?	0 TO 100%
GROUP WORK	Does the student demonstrate the ability to explain how he/she worked and performed in group projects?	0 TO 100%
PRESENTATION LAYOUT		0 TO 100%

	Does the presentation of the student look professional?	
TOTAL		**100**

8.0 : REPORT

DURATION OF TRAINING GUIDELINE	1 MONTH
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	5.00%
CREDIT VALUE	3
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

GUIDELINE AND INSTRUCTIONS TO STUDENT

It is essential that the workbook reports should contain the detail of all tasks allocated to the student during the months of Work Integrated Learning. Reports written in "telegram style" will not be acceptable. These reports should contain enough data to place academic staff in a position to evaluate the Work Integrated Learning done as either acceptable or unsatisfactory for the specific period.

REPORT

The students are required to compile a report regarding the work he/she performs and submit it via the TUT MyWIL portal.

The report must be compiled in the following manner:

Introduction (50 words)

This section helps the reader understand your workplace and must include:

- **The geographical location of your workplace;**
- **The name of your employing organization;**
- **The position occupied by you;**
- **The chronology-duration of your work with the organization.**

Background (300 words)

This section provides the context in which you are working. It should include details such as:

- **The nature of the work performed by your section/department;**
- **The department's objectives;**
- **The nature of your work area;**
- **A statement of your duties;**

Personal Role at Workplace (1000-2000 words)

This section must describe the actual work performed by you. If you work in a team, then it is not sufficient to narrate the work performed by the team; your own role must be clearly stated. You will be

evaluated based on the competencies that you claim in this section. Include details such as:

- **Technical details of your work;**
- **How you have applied your knowledge and skills;**
- **The specific duties delegated to you and how you accomplished them;**
- **How you worked with the other team members (if applicable)?**

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA		TYPE
COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

9.0 : MENTOR EVALAUTION

DURATION OF TRAINING GUIDELINE	6 MONTHS
CORE OR ELECTIVE	CORE
WEIGHT OF OUTCOME	25.00%
CREDIT VALUE	15
ACADEMIC	50%
MENTOR	50%

OUTCOME DETAILS

GUIDELINES AND INSTRUCTIONS TO STUDENT

The student is required to submit the "**MENTOR EVALUATION FORM**", This form is accessible through the TUT MyWIL portal's Documents Menu, located under the Forms and Templates section. Ensure that the Mentor evaluation form is duly completed and signed by your Industry Mentor before submission.

ASSESSMENT RUBRIC

ASSESSMENT CRITERIA		TYPE
COMPETENCY LEVEL	Rate the student as Competent (100%) or Not Competent (0%)	COMPETENT / NOT COMPETENT
TOTAL		**COMPETENT / NOT COMPETENT**

FORMS AND TEMPLATES

TITLE
LOGSHEET
TECHNICAL EVALUATION
MENTOR PACKAGE